



Shenzhen Guanglijin Technology Co., Ltd.
SHENZHEN GUANGLIJIN TECHNOLOGY CO., LTD.

5-350Jewelry machine instruction manual



Table of contents

1. Safety Instructions

2. Machine function and parameters

3. Function introduction of CNC system operation panel

4. Common programmingG , Mthe code

5. Maintenance and maintenance of machine tools

6. Common troubleshooting

7. Common settings and operation methods

8. Accessories

Nine. Electrical schematic diagram

1. Safety Instructions

Safety Precautions:

Before installation, power on, operation, maintenance and inspection, you must be familiar with the terms in this manual and all relevant safety precautions to ensure correct use.



Please confirm the strength of the ground.

Make sure the machine is securely mounted and in the correct working position.

Do not let water, yarn ends, wood chips, dust, metal and other foreign objects fall into the electric control box. Failure to do so may cause a fire or accident.

Notice!

When moving the device, please use the correct lifting tools to prevent the device from tilting and falling, or from strong impact.



The power cord and connectors must be connected after the equipment is installed.

Otherwise, electric shock or injury may occur.

The power cord must be connected only after the main switch is disconnected.

The wiring operation must be performed by a professional electrician, otherwise electric shock accidents may occur. A reliable grounding wire must be used alone, and the grounding resistance is less than 10Ω .

Otherwise, electric shock or fire accident may occur.

The machine driving voltage is 380V, Non-professionals are not allowed to disassemble, refit, or repair. It is strictly forbidden to adjust the touch when the machine is running, otherwise there is a danger of being involved in cutting off. When the machine is running, it is not allowed to check the surface finish of the workpiece and measure the size of the workpiece by hand.



Before the machine is running, check whether the fastening screws are fastened reliably.

Ensure that the air source is dry and the air pressure is stable.

Pay attention to ensure the air pressure of the spindle air seal, otherwise it will cause the handle to loosen. Tools, machined parts must be clamped.

Keep work area clean and dry.

Do not open the electric control box after the equipment is powered on.

After starting up, it needs to return to the origin of the machine tool.

The operator's head, hands, body, etc. enter the machine for cleaning and inspection, and must press the emergency stop switch before proceeding.

Prevent electric shock accident or personal injury.

Notice!

Before running the machine, please confirm the distance between the tool and the machined part, and the cutting depth. Do not plug or unplug any cables while the power is on.

Do not operate the power switch continuously for a short period of time.

Otherwise equipment damage may occur.



Before making any adjustments to the machine, turn off the power and let the machine stop. When checking and maintaining the electrical part, it must be carried out by a professional electrician.

Please disconnect the external input power before maintenance.

The inspection and maintenance of the spindle drive part must be done after disconnecting the power supply. 5 minutes or more. Otherwise personal injury or electric shock may occur.

Security!

- 1.This machine tool is a precision machine, and it has been debugged before leaving the factory. Please do not modify the system parameters at will, and it is forbidden to knock on the machine tool body.
- 2.Please designate professionals for training, maintenance and operation of equipment.
- 3.When the machine is working, please close the protective door to prevent the cutting fluid from splashing.
- 4.Find problems and solve them in time, and prohibit the use of equipment for sick operations.
- 5.When working with external equipment such as oil pumps, pressure air pumps, and vacuum pumps, please ensure safety and reliability.
- 6.It is necessary to wear labor protection supplies before operation to ensure personal safety and health.

2. Machine function and parameters

The machine model is:GLJ 5- 350V15 (Five-axis), the transmission part adopts high-precision screw, guide rail and high-end servo drive system, while using intelligent programming software to make it

The operation speed is faster, the effect is good, the special process is flexible, and the research and development space is larger.

The machine parameters are shown in the table below:

序号	项目	参数
1	主轴转速	3.7KW/40000rpm
2	伺服马达功率	750W/400W
3	刀柄规格	ISO20
4	夹头规格	ER16
5	刀具装夹范围	Φ 1-8mm
6	重复定位精度	± 0.005mm
7	XYZ轴运行往复精度	± 0.01/300mm
8	工作平面度	± 0.015mm
9	XYZ行程	350*220*185
10	机床外观尺寸	1200*1480*2000
11	机身重量	1.7吨
12	电力配置	三相四线 380V/50HZ

3. Function introduction of CNC system operation panel



7# 如果需要输入符号 '#', 需要先按一下 (shift), 再按 .

重置
可重置机床状态
可终止程序、转速等

程序编辑
按下此键可直接跳转至
'程序编辑' 页面

加工监控
按下此键可直接跳转至
'加工监控' 页面

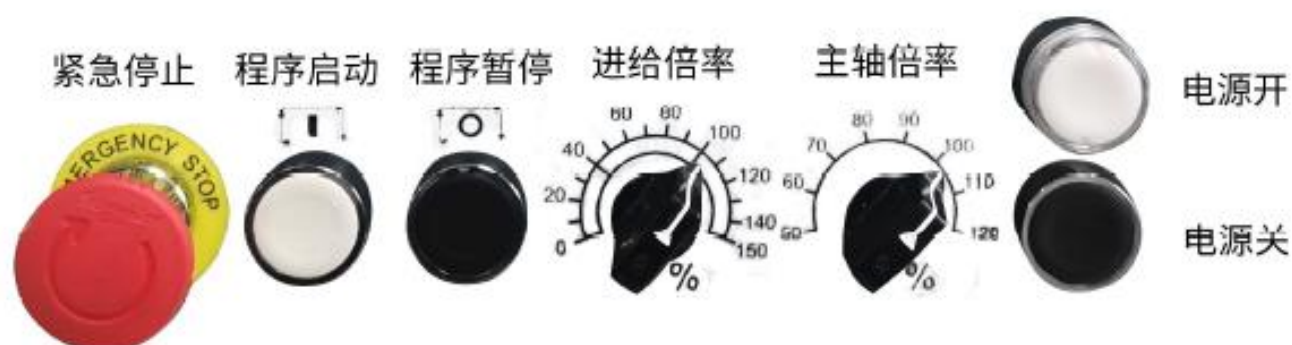
如果需要输入字母 'K', 快速按两次 即可, 或者先按 (shift), 再按 .

教导
用以辅助输入当前坐标
或参考当前位置创建简单程序

帮助
可显示帮助文件
G 代码说明表等信息

上一页 下一页

五轴机 新代数控系统操作面板图释



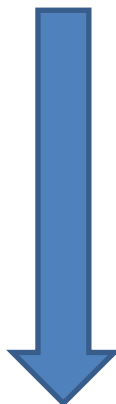
	返回参考点模式	在此模式按下‘程序启动’各轴皆返回参考点 在此模式按 X+、B+、W+ 等按键，则是单轴返回参考点
	手轮模式	启用此模式，可通过 MPG(手轮)控制机床各轴移动。
	手动连动模式	启用此模式，按 X+、X-、Y+、Y-、B+、W- 等按键可使主轴（工作台）连续移动。 移动速度由进给倍率旋钮决定
	手动寸动模式	启用此模式，按 X+、X-、Y+、Y-、B+、W- 等按键可使主轴（工作台）移动指定距离， 移动距离由 G00 倍率决定。25%=0.01mm 50%=0.1mm 100%=1.0mm
	快速连动	在手动连动模式，按住此键 再按 X+、X-、Y+、Y-、B+、W- 等按键 可使主轴（工作台）快速移动。
	MDI 模式	在此模式，可在机床上手工输入程序，输入程序会存储在 MDIBlock 里 当按下‘程序启动’键执行的是 MDIBlock 里的程序，而不是 NC 文件里的程序
	自动执行模式	在此模式按下‘程序启动’键，数控系统会自动执行当前 NC 文件里的程序
	手轮试切模式	在‘自动执行’和‘MDI’模式下，启用‘手轮试切模式’，程序将不会自动执行 当转动手轮时，程序执行，停止转动手轮，则暂停执行程序
	单节执行模式	在‘自动执行’和‘MDI’模式下，启用‘单节执行’模式， 每按一次‘程序启动’键程序执行一行
	选择性跳跃模式	在‘自动执行’和‘MDI’模式下，启用‘选择性跳跃’模式， 在‘/’之后的程序将跳过不执行，否则执行
	选择性程序暂停	在‘自动执行’和‘MDI’模式下，启用‘选择性程序暂停’模式， 在程序执行到 M01 时，程序暂停，否则程序不暂停。

4. Common programming G , M the code

G00	直线快速定位	G73	高速啄式钻孔循环
G01	直线插补	G81	钻孔循环
G02	顺时针圆弧插补	G82	孔底暂停钻孔循环
G03	逆时针圆弧插补	G83	啄式钻孔循环
G70	英制 (G20)	G80	取消钻孔循环
G71	公制 (G21)	G15	启用极坐标
G90	绝对模式	G16	取消极坐标
G91	增量模式	G68	启用坐标旋转
G04 X_	暂停, 例如: G04 X2. 暂停两秒	G69	取消坐标旋转
G52	局部坐标设定 (坐标偏移)	F_	进给速度, 例如: F1500 表示进给速度为每分钟 1500mm (公制)
G28	从指定点返回参考点	S_	主轴转速, 例如: S6000 M03 主轴以每分钟 6000 转的速度正转
G54	第一工作坐标系	M03	主轴正转 (顺时针)
G55	第二工作坐标系	M04	主轴反转 (逆时针)
G56	第三工作坐标系	M05	主轴停止
G57	第四工作坐标系	M19	主轴定位
G58	第五工作坐标系	M08	开启冷却液
G59	第六工作坐标系	M09	关闭冷却液
G17	设定 X-Y 工作平面	T_	刀号, 例如: T6 表示 6 号刀
G18	设定 Z-X 工作平面	M06	换刀, 例如: T5 M06 换 5 号刀
G19	设定 Y-Z 工作平面	M16 C_	圆等分循环, 例如: M16 C80. C 轴等分为 80 份
G43	调用刀长补偿	M61	取消圆等分循环
H_	刀长补偿号, 例如: G43 H2 调用 2 号刀长补偿	M62	取消圆等分循环
G49	取消刀长补偿	M00	程序暂停
G41	刀具半径左补偿	M01	程序选择性暂停
G42	刀具半径右补偿	M30	程序结束
D	半径补偿号, 例如: G41 D3 调用 3 号半径补偿为左补偿	M98	调用子程序
G40	取消半径补偿	M99	返回主程序

5. Maintenance and maintenance of machine tools

- 1.This machine tool is equipped with an automatic lubrication system. The oil injection pump is installed in the machine cover on the right side of the machine. It will automatically and regularly lubricate the guide rails and screw rods of the machine tool. When the oil is insufficient, the system will automatically alarm and remind you to refuel. Please use qualified guide oil.
- 2.When not in use for a long time, please run with oil regularly, fill oil first and let it stand30Minutes later, the machine will run again.
- 3.After processing, the chips and other sundries of the machine tool must be cleaned immediately, and guide rail oil should be sprayed on the exposed surfaces such as guide rails and workbenches for anti-rust protection.
- 4.Maintenance of the water cooler: The maintenance of the cooler is crucial to the normal operation of the spindle. For the maintenance method, see the "Maintenance and Maintenance" posted on the side of the chassis. In order to extend the service life of the spindle, it is recommended that customers useBOSCHBrand "Bosch Motor Vehicle Engine Coolant" instead of purified water.
- 5.The oil should not exceed the limit mark of the water cooler. When it is lower than the limit mark of the water cooler, the oil needs to be replaced in time, and it is recommended to replace the oil every two years.
- 6.The exhaust fan of the hood electrical box needs to replace the filter cotton regularly to prevent the cutting oil in the air from penetrating into the chassis and affecting the electrical components.
- 7.The exhaust fan of the machine control box needs to replace the filter cotton regularly to prevent the cutting oil in the air from penetrating into the chassis and affecting the control system.
- 8.Check oil supply pipes and connections daily for oil leakage and maintain in time. Good lubricating effect can take away the heat, reduce the working temperature of the bearing and prolong the service life.
- 9.The following are pictorial descriptions of several specific maintenance locations:



9.1 Replace the filter cotton for the large fan on the hood

9.1.1 Each hood has 4 big fans.

9.1.2 If the filter cotton is not cleaned for a long time, the cutting oil in the air will penetrate into the chassis and affect the electrical components.

9.1.3 The filter cotton needs to be replaced regularly when the equipment is in use.

9.1.4 The specific operation steps are shown in the figure below.



4个大风扇



拆保护盖



换过滤棉



完成

9.2Control box small fan replacement filter cotton

9.2.1Each hood has2a small fan.

9.2.2If the filter cotton is not cleaned for a long time, the cutting oil in the air will penetrate into the control box and affect the control system.

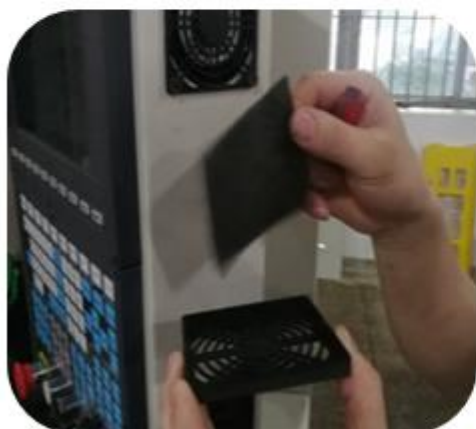
9.2.3The specific operation steps are shown in the figure below.



2个小风扇



拆保护盖



换过滤棉



完成

9.3 Oil mist separator replacement filter cotton

9.3.1 Each oil mist separator has 2 layer of filter cotton.

9.3.2 If the filter cotton is not cleaned for a long time, the cutting oil will flow back into the large hose of the oil inlet, which will also affect the oil mist separation effect.

9.3.3 The specific operation steps are shown in the figure below.



拆3个钣金固定旋钮



2层过滤棉



换2层过滤棉



完成

9.4 Maintenance of water cooler (see picture below)

9.4.1 The maintenance of the cooling machine is very important to the normal operation of the spindle and needs to be maintained carefully.

9.4.2 For the maintenance method of the cooling machine, see "Maintenance and Maintenance" posted on the side of the chassis.

9.4.3 In order to extend the service life of the spindle, it is recommended that customers use BOSCH Brand "Bosch Motor Vehicle Engine Coolant" instead of pure water.

9.4.4 The dust-proof net in front of the water chiller needs to be cleaned regularly to avoid affecting the cooling effect.



9.5 Air source system

A clean and dry air source is very important to the use of the equipment, it is recommended to use the air supply installation as shown in the figure below.



6. Common troubleshooting

Alarm No. number / report police name	reason	confirmation method	Treatment measures
A.400: Overvoltage (servo unit interior master loop power checked out Voltage)	The power supply voltage exceeds the cell range	Measure the supply voltage.	WillAC/DCThe power supply voltage is adjusted to be within the product specification range.
	Power is unstable status, or subject to The effects of lightning strikes	Measure the supply voltage.	Improve your power situation, set up a surge suppressor Then turn on the power supply of the servo unit again. still When an alarm occurs, it may be the servo unit Fault. Replace the servo unit.
	ACPower supply voltage exceeds When exceeding the specification range, advise Acceleration and deceleration	Determine the supply voltage and running speed, torque.	WillACSupply voltage is regulated to specification range Inside.
	External regenerative resistor value greater than the operating conditions	Confirm operating conditions and Regenerative resistance value.	Taking into account operating conditions and loads, revisit Regenerative resistance value.
	in allowable moment of inertia ratio or mass ratio run in the state of	Confirm moment of inertia ratio or mass ratio in the allowable within the range.	Extend the deceleration time, or reduce the load.
	Servo unit failure	- - -	When the main circuit power supply is not connected, Turn on the control power of the servo unit again, When the alarm still occurs, it may be the servo Unit failure. Replace the servo unit.

Alarm No. number / report police name	reason	confirmation method	Treatment measures
A.710: overload (momentary most heavy load) A.720: overload (even continued max)	Motor wiring, coding The wiring of the device is bad or the Poor connection	Confirm the wiring.	Check the wiring of the motor and the wiring of the encoder something wrong.
	The motor runs beyond the Overload Protection Features	Confirm motor overload Features and Run Instructions	Re-discuss load conditions, operating conditions. Or revisit the motor capacity.
	due to mechanical factors resulting in the motor not driving move, causing runtime The load is too large	Confirm the run command and motor speed.	Improve the mechanical factor.
	Optical Linear Encoder Scale pitch (Pn282) The setting is abnormal	confirmPn282set Value.	WillPn282Set to an appropriate value.
	Motor phase sequence selection (Pn080=n.□□ X) Abnormal	confirm Pn080=n.□□X □ set value.	WillPn080=n.□□X□Set as appropriate value.
	Servo unit failure	- - -	It may be that the servo unit is faulty, replace the servo unit service unit.
A.7Ab: Servo unit built-in fan stop	inside the servo unit fan stops	Confirm whether it is stuck foreign body.	If the alarm still occurs after removing the foreign matter, It is possible that the servo unit is faulty. replace the servo service unit.

Alarm No. number / report police name	reason	confirmation method	Treatment measures
A.810: Encoder alert	Absolutely the first time Power supply for value encoders	Confirm whether it is the first Turn on the power for the first time.	Perform encoder setting operations.
	Remove the encoder cable later connected	Confirm whether it is the first Turn on the power for the first time.	Confirm the connection of the encoder and perform the encoder setting operation.
	Servo unit control power supply (+5V)by and battery power break down	Confirm encoder connection battery and connections to the Is the device status correct? •	Restore power to encoder (replace battery, etc.) After that, perform the setting operation of the encoder.
	Absolute encoder failure barrier	- - -	It cannot be solved even if the setting operation is performed again. When the alarm is cleared, replace the servo motor.
	Servo unit failure	- - -	It may be that the servo unit is faulty, replace the servo unit service unit.
A.820: encoder and Calibration Alert	Encoder failure	- - -	■ Absolute encoder Reset the encoder. still frequently When it happens, the servo motor may be faulty. Replace the servo motor. ■ 1 rotary absolute encoder or incremental Quantitative encoder * It may be the fault of the servo motor, Replace the servo motor. * The linear encoder may be malfunctioning. Replace the linear encoder.
	Servo unit failure	- - -	It may be that the servo unit is faulty, replace the servo unit service unit.

Alarm No. number / report police name	reason	confirmation method	Treatment measures
A.840: Number of encoders According to the alert	Encoder malfunction	- - -	Reconnect the cables of the servo unit. still When the alarm occurs, it may be the servo motor Or the linear encoder is faulty. Replace the servo motor machine or linear encoder.
	Linear Encoder Read take error	- - -	Linear encoders are not mounted with reasonable tolerances. Reinstall the linear encoder.
	Linear Encoder Speed is too big	- - -	Reduce motor speed to linear encoder production Below the speed specified by the manufacturer, turn on the control power supply.
	Interference due to noise resulting in encoder error action	- - -	Correctly carry out the wiring around the encoder (point Encoder cable and servo motor main circuit cables, grounding treatments, etc.).
	connection of magnetic pole sensor incorrect line	Confirm magnetic pole sensor wiring.	Correct the wiring of the magnetic pole sensor
	Pole sensor failure	- - -	Replace the magnetic pole sensor.

Alarm No. number / report police name	reason	confirmation method	Treatment measures
A.C90: encoder pass Communication failure	Encoder connector Bad contact or wiring mistake	Confirm that the encoder is connected with status of the connector.	Insert the encoder connector again, and confirm Encoder wiring.
	Encoder cable disconnected , short circuit, or using a exceeding the specified impedance cable	Confirm the power consumption of the encoder condition of the cable.	Use an encoder cable of the specified specification.
	temperature, humidity, air Corrosion caused by body; water droplets, cutting oil caused by short circuit; vibration caused by the connector Poor touch	Confirm the usage environment.	Improve the use environment and replace the cable. even though If this still does not improve, replace the servo service unit.
	produced by noise disturbance Make a mistake	- - -	Correctly carry out the wiring around the encoder (point Encoder cable and servo motor main circuit cables, grounding treatments, etc.).
	Servo unit failure	- - -	Connect the servo motor to other servo units When the control power is turned on, if there is no alarm, it may be the fault of the servo unit barrier. Replace the servo unit.

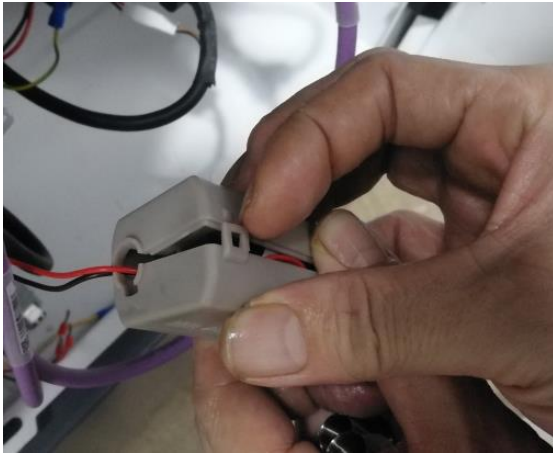
Alarm No. number / report police name	reason	confirmation method	Treatment measures
A.910: overload (become over load alert (A.710or A.720)Of previous warning)	Motor wiring, coding The wiring of the device is bad or the Poor connection	Confirm the wiring.	Check the wiring of the motor and the wiring of the encoder something wrong.
	The motor runs beyond the Overload Protection Features	Confirm motor overload Features and Run Instructions	Re-discuss load conditions, operating conditions. Or re-discuss the motor capacity.
	due to mechanical factors resulting in motor failure move, causing runtime The load is too large	Confirm the run command and motor speed.	Improve the mechanical factor.
	Overload warning value (Pn52B) discomfort when	Confirm overload warning value (Pn52B)whether appropriate	Appropriately set the overload warning value (Pn52B).
	Servo unit failure	---	It may be that the servo unit is faulty, replace the servo unit service unit.

7. Common settings and operation methods

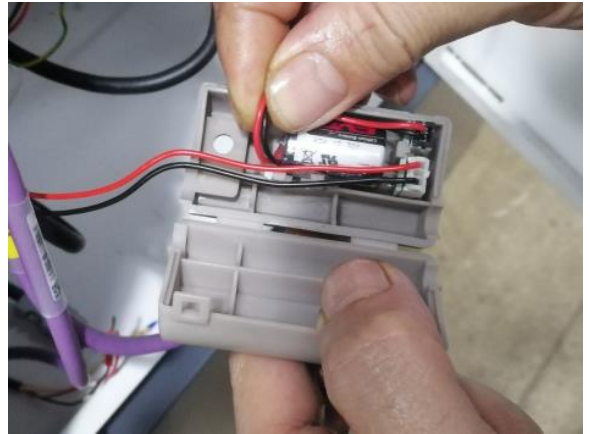
During the use of the machine tool, when the machine tool alarms "810", it means that the driver battery is out of power. At this time, it is necessary to replace the driver battery and reset the "mechanical origin coordinates" position. The specific steps are as follows:

1) Battery replacement steps:

- 1.1) Press the emergency stop button;
- 1.2) Press the power off button

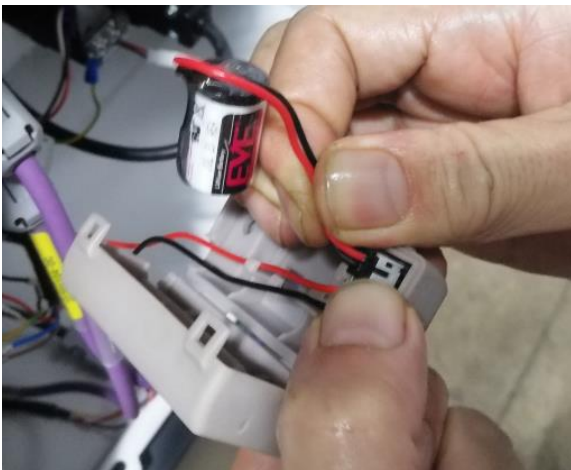


- 1.3) Unplug the side of the battery box
two snap buttons

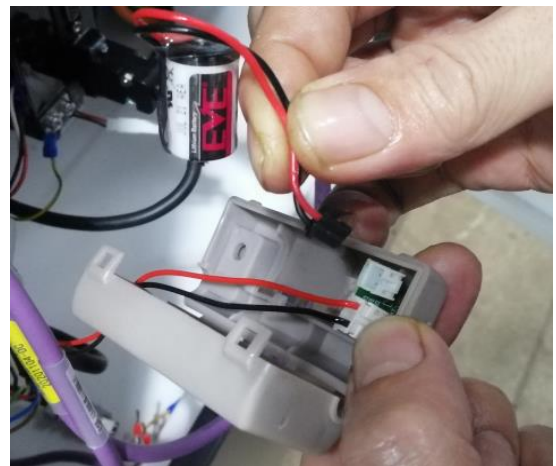


- 1.4) Unplug the battery cable

remove the battery



- 1.5) upwards and unplug the battery
Terminals



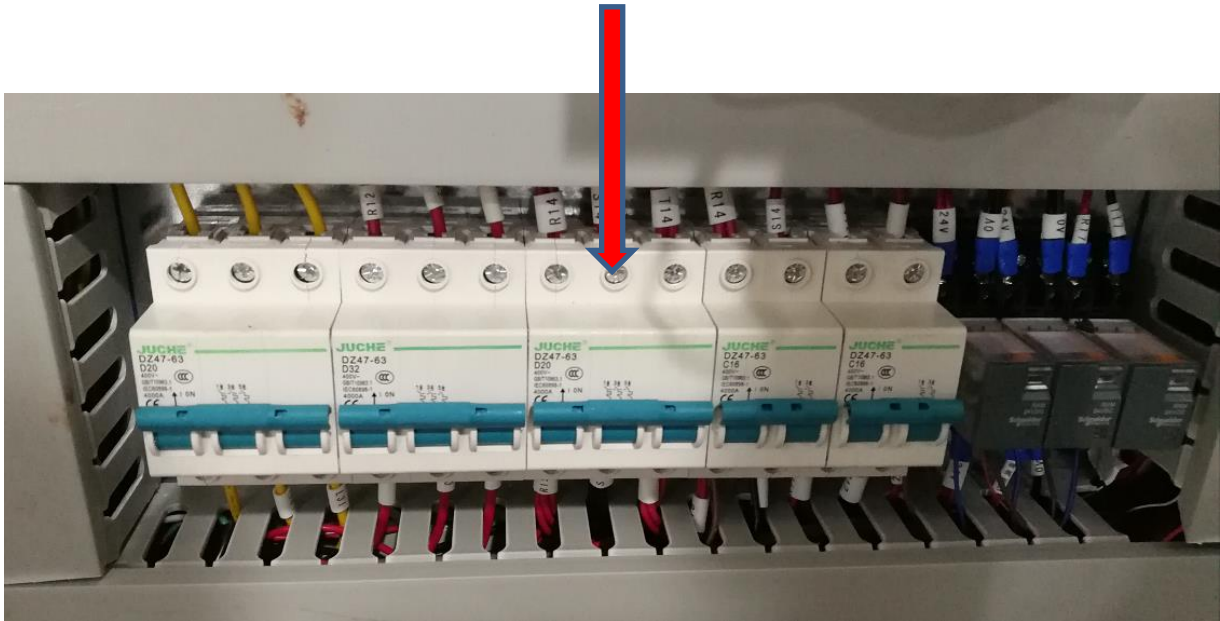
- 1.6) reverse operation into the
new battery

1.7) Press the power button;

1.8) to be displayed on the touch screen 100% After fully powered on, open the electrical box and disconnect the circuit breaker connected to the drive (as shown below)

(The purpose of this step is: to reset the motor encoder)

1.8),1.9) step open and close the third circuit breaker



1.9) re-closed "1.8)" step disconnected circuit breaker;

1.10) Press the power off button;

1.11) Press the power button to restart;

1.12) Release the emergency stop switch.

2) Set the main axis X, Y Origin mechanical coordinate steps:

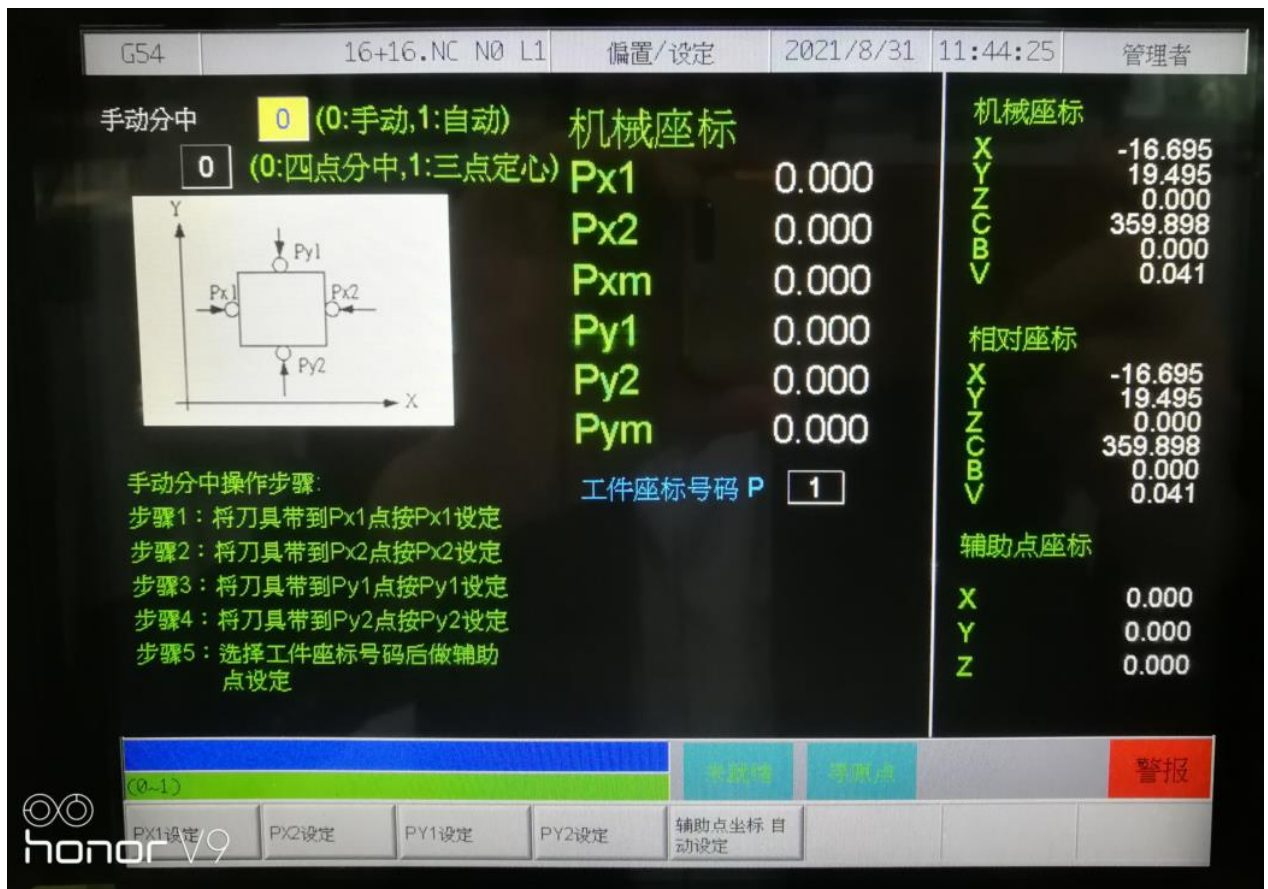
2.1) Mounting jig, copper blank, $\phi 6$ knives;

2.2) to set the spindle speed as: 12000rpm;

2.3) Use the handwheel to move the spindle, and align the tool to the center of the copper embryo (closer);

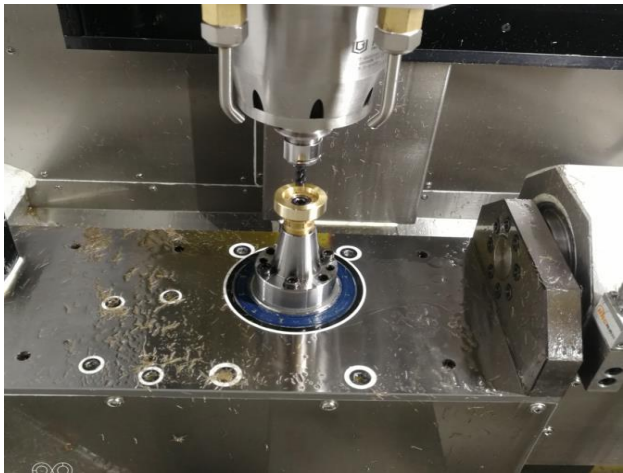
2.4) to turn the screen to the "Relative Coordinates" page, and press the "Relative Coordinates All Clear" button;

2.5) to convert the control panel to the following page:



Manual split screen

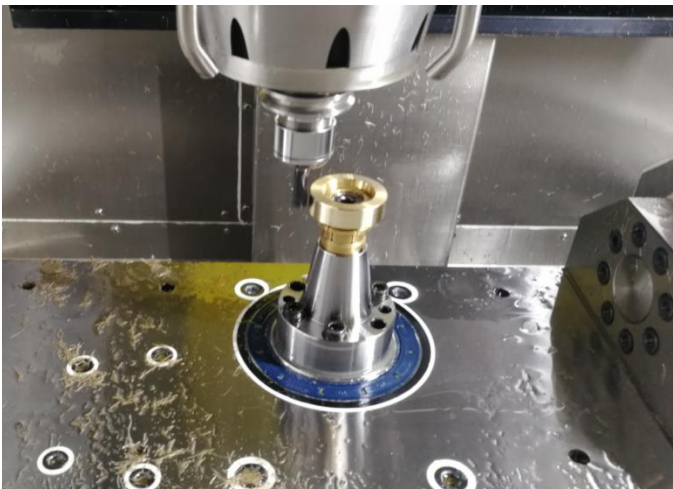
2.6) to turn on the spindle, along YSlowly move the spindle tool to the copper blank in the axial direction12point position, handwheel shake4axis, coarse, fine The outer circle of the gong copper embryo, pressF3key, The screen is now complete"Py1set up"



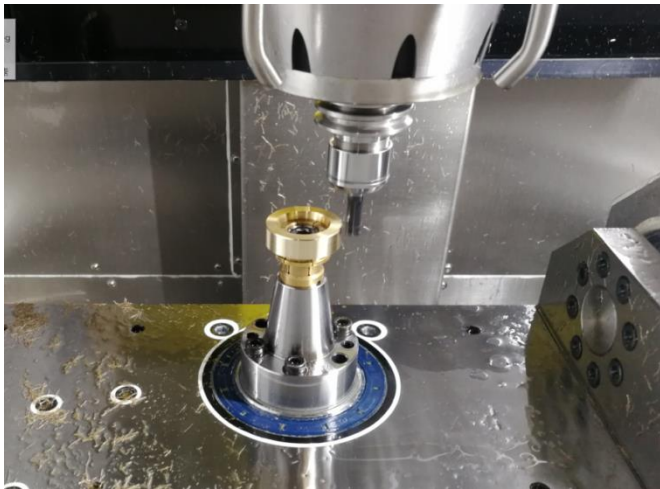
2.7) liftZaxis,xThe position of the axis remains the same, along the YSlowly move the main shaft to the copper blank in the axial direction 6Point position, use the rotating tool to slowly Close to the outer circle of the copper embryo, make sure to press theF4 key, and the screen is complete"Py2set up" and will automatically derive"Pym"data.



2.8) liftZAxis, move the main axis to "relative coordinates" Yaxis"0"point position along xSlowly move the main shaft to the copper blank in the axial direction 9Point position, use the rotating tool to slowly Close to the outer circle of the copper embryo, make sure to press theF1 key, and the screen is complete"Px1set up"



2.9) liftZaxis,YThe position of the axis remains the same, along the x Slowly move the main shaft to the copper blank in the axial direction 3Point position, with the rotating tool slowly close Close to the outer circle of the copper embryo, make sure to press theF2 key, and the screen is complete"Px2set up" and will automatically derive"Pxm" data.



2.10) At this time, the

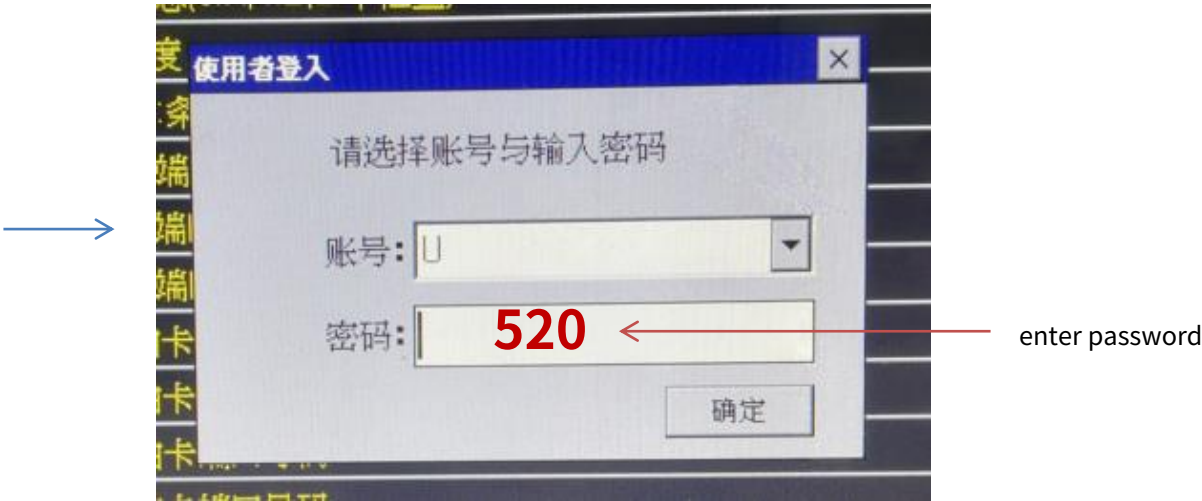
PxmThe value ----- is the machine tool "xaxis mechanical coordinate origin" Pym

The value ----- is the machine tool "Yaxis mechanical coordinate origin"

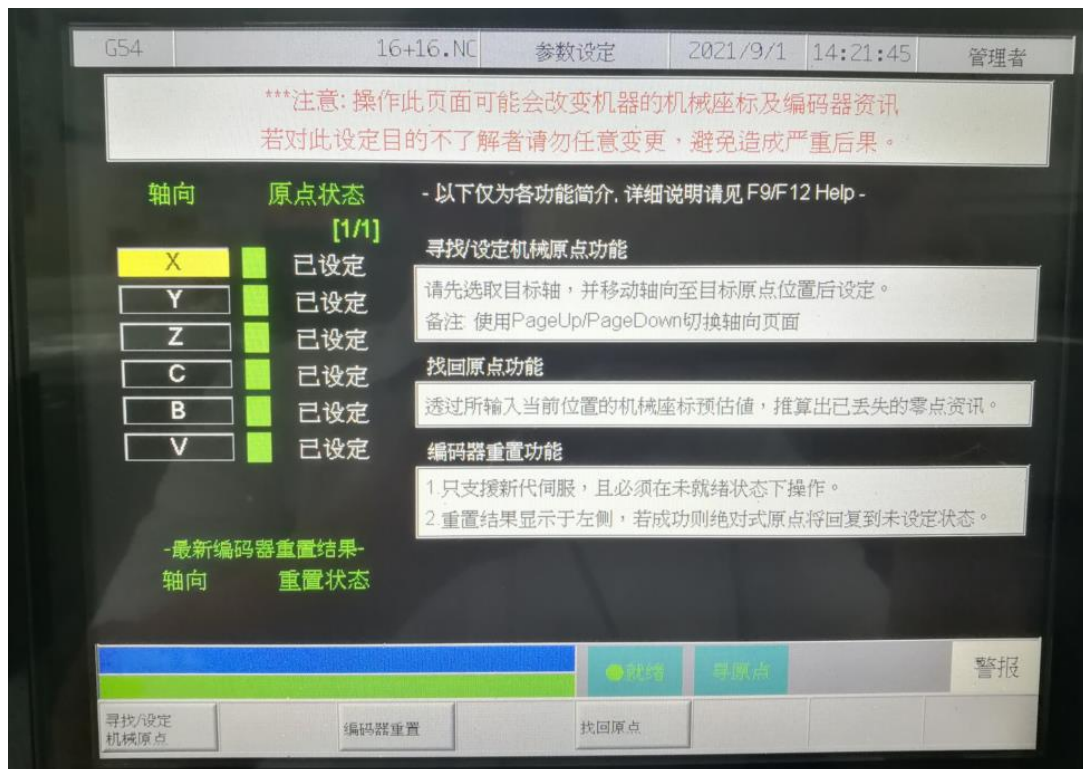
2.11) In the "Mechanical Coordinate" system at the upper right of the "Manual Split Center Screen", use the handwheel to move the spindle to "Pxmvalue of " and "PymThe position of the value of ", the position of the spindle is "xaxis,YAxis machine coordinate origin" position.

2.12) on the control screen and press  key to turn on the light, and enter the "return-to-origin mode".

2.13) On the main page of "Mechanical Coordinates", select  



xcoordinate
origin



Move the yellow marker to the image above "X" Location

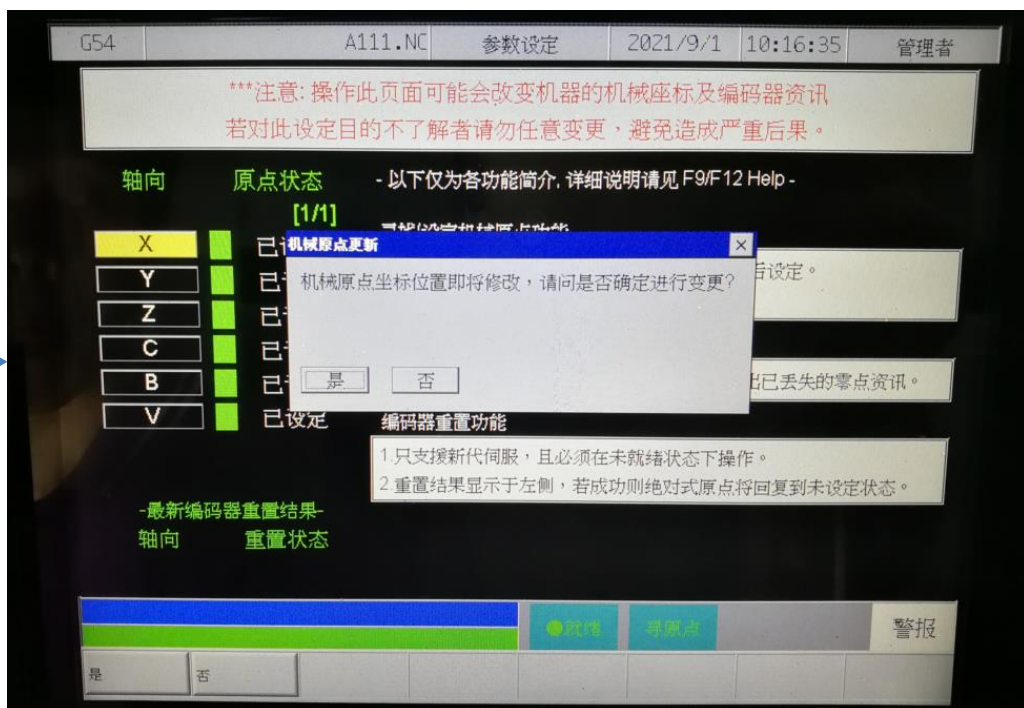
choose

find/set

Mechanical origin

The screen displays as follows:

xcoordinate
origin



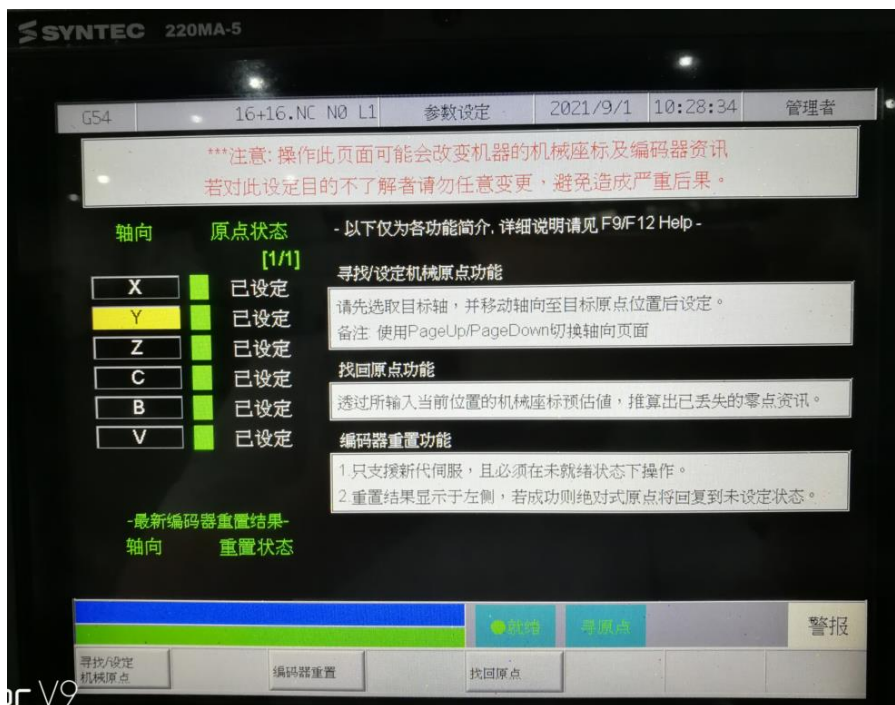
choose

Yes

The origin of the axis mechanical coordinates is set

Continued from previous page, set "Y" Axis mechanical coordinate origin:

Ycoordinate
origin



Move the yellow marker to the image above "Y" Location

choose

find/set

Mechanical origin

A jump window appears (omitted)

choose

Yes

The axis mechanical coordinate origin setting is completed.

At this time, the main page of "Mechanical Coordinate" displays "X" "Y" The coordinates are:0.000, as shown in the figure



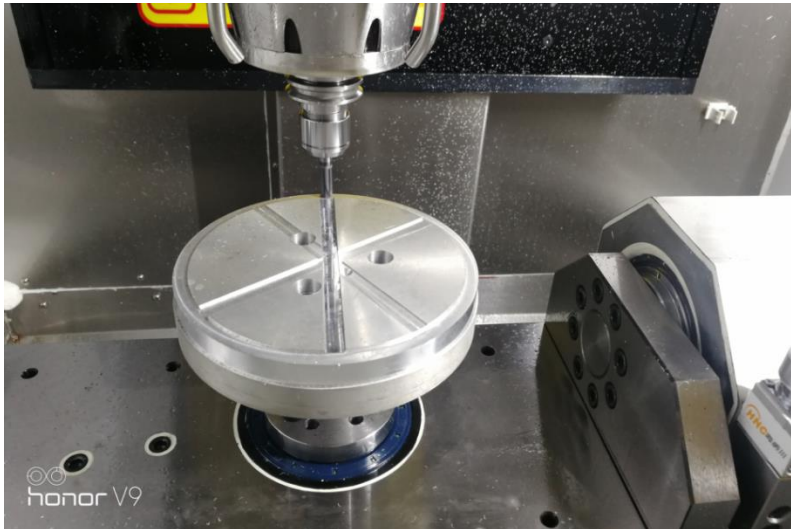
3) Settings B Axis (five-axis) origin mechanical coordinate steps:

3.1) mounting base, $\phi 100\text{mm}$ Aluminum plate, $\phi 6$ knives;

3.2) to set the spindle speed as: 18000rpm;

3.3) restore the mechanical coordinates of the machine tool to the origin position;

3.4) will be the main axis along the Y Shaft moves to aluminum disc 12 At the point position, on the surface of the aluminum plate, a rough and fine surface pit is formed (as shown in the figure 6H—12H direction surface pit), see the figure below:



3.5) to turn the screen to the "Relative Coordinates" page, and press the "Relative Coordinates All Clear" button;

3.6) by turning the handwheel to C axis rotation 180° , at this point, the page is displayed as shown below:



3.7) along with the dial indicatorYThe beating of the plane of the gong and the pit in the axial direction, as shown in the figure below:



3.8) as the dial indicator jumps at $0 \sim 0.01\text{mm}$ between, it can be determined that the current position is B Axis (five-axis) mechanical coordinate origin position.

3.9) such as the dial gauge jumps $>0.01\text{ mm}$, you need to reset the B Axis origin machine coordinates. At this time, fine-tune the five-axis, alongYThe shaft moves the dial indicator to check the plane drawn above until it is leveled and there is no beating. At this time, the page is displayed as follows:



E.g:
Normal five-axis seat
marked as:0.010

3.10) turning the handwheel will BShake the coordinates of the axis to half of the page display, as shown in the figure above: 0.010, shake to 0.005 location; if so: 0.009, then shake to 0.004 location; and so on...

at this time 0.005 or 0.004 The location is B The new origin position of the axis (five-axis) mechanical coordinates needs to be reset in the next step.

3.11) followed by "set the spindle X, Y Origin machine coordinate step: "2.12, 2.13 In the same way, reset B Axis (five-axis) origin machine coordinates.

3.12) check B Whether the axis origin coordinates are correct. repeat 3.4 ~ 3.8 Step action check.

3.13) Remarks: If there is no ϕ in this step 100mm Aluminum plate, you can directly use the dial indicator to set the surface of the divider to set the coordinates of the five-axis origin.

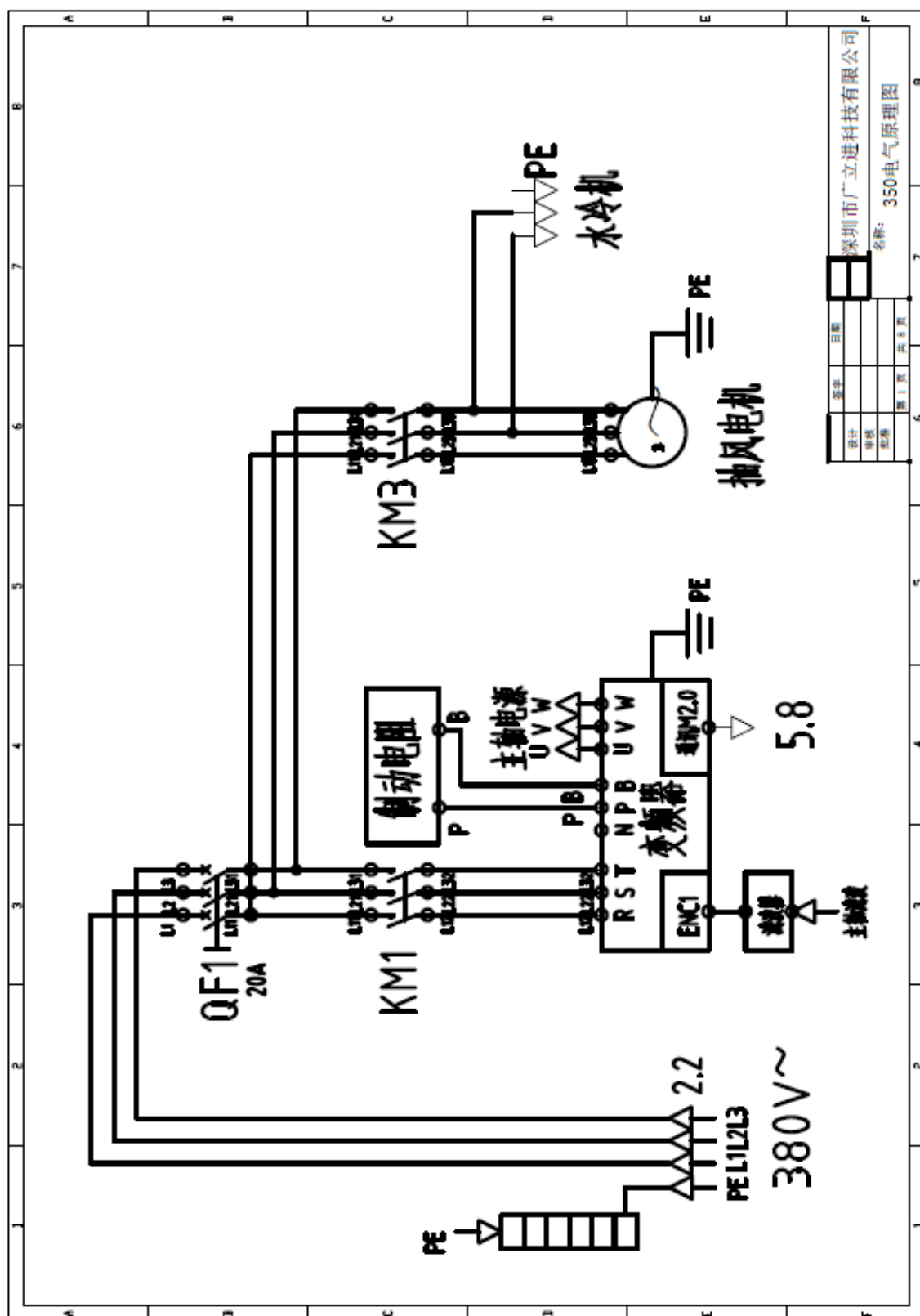
3.14 Special Note: Reset the coordinates of the five-axis mechanical origin
back, Y The mechanical origin coordinates of the axis will change,
Need to reset Y Axis mechanical origin coordinates!
If set after Y Axis coordinates compared with original coordinates
for +0.200mm above, you need to reset Y Axis tool
change point coordinates.

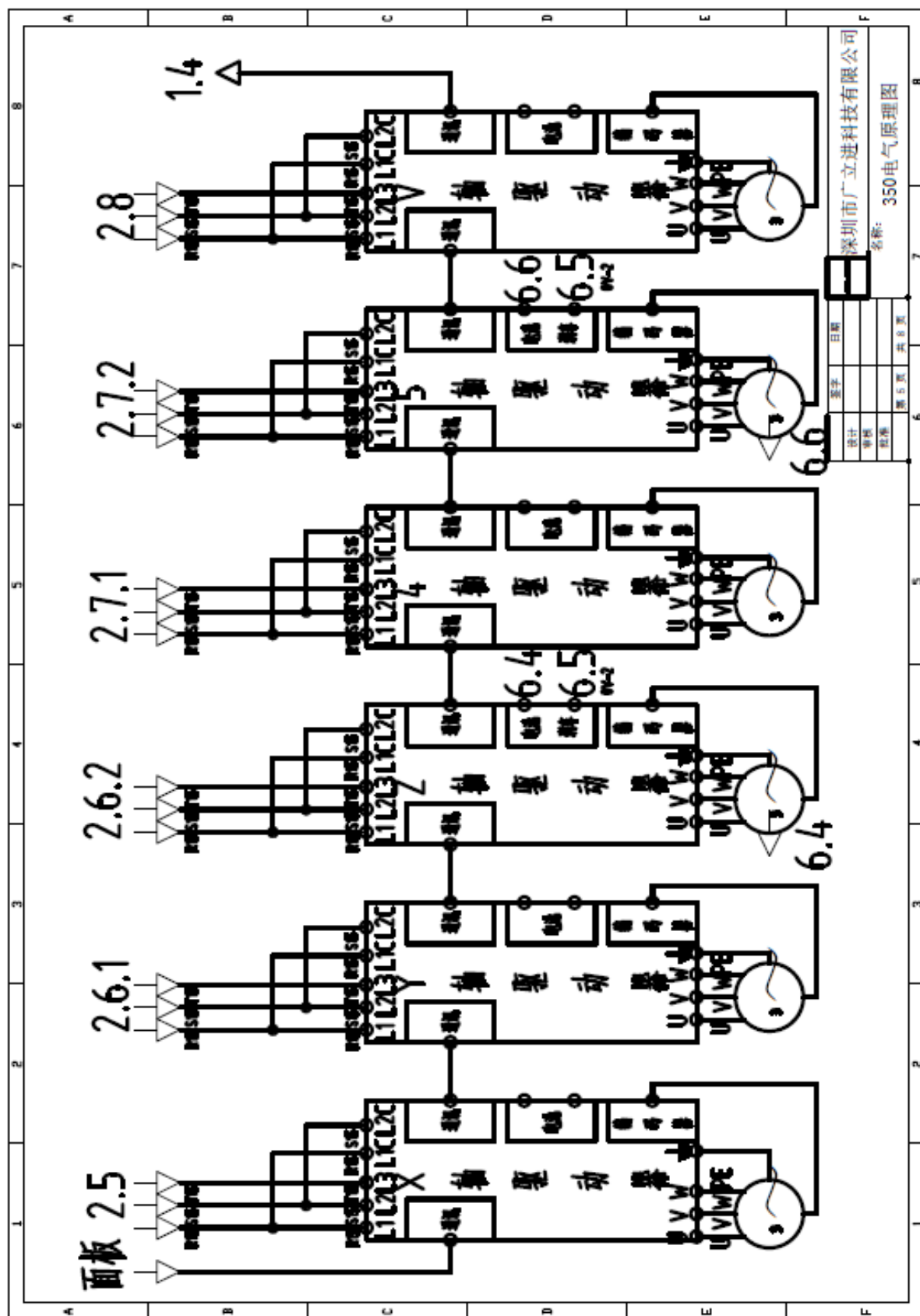
8. Accessories

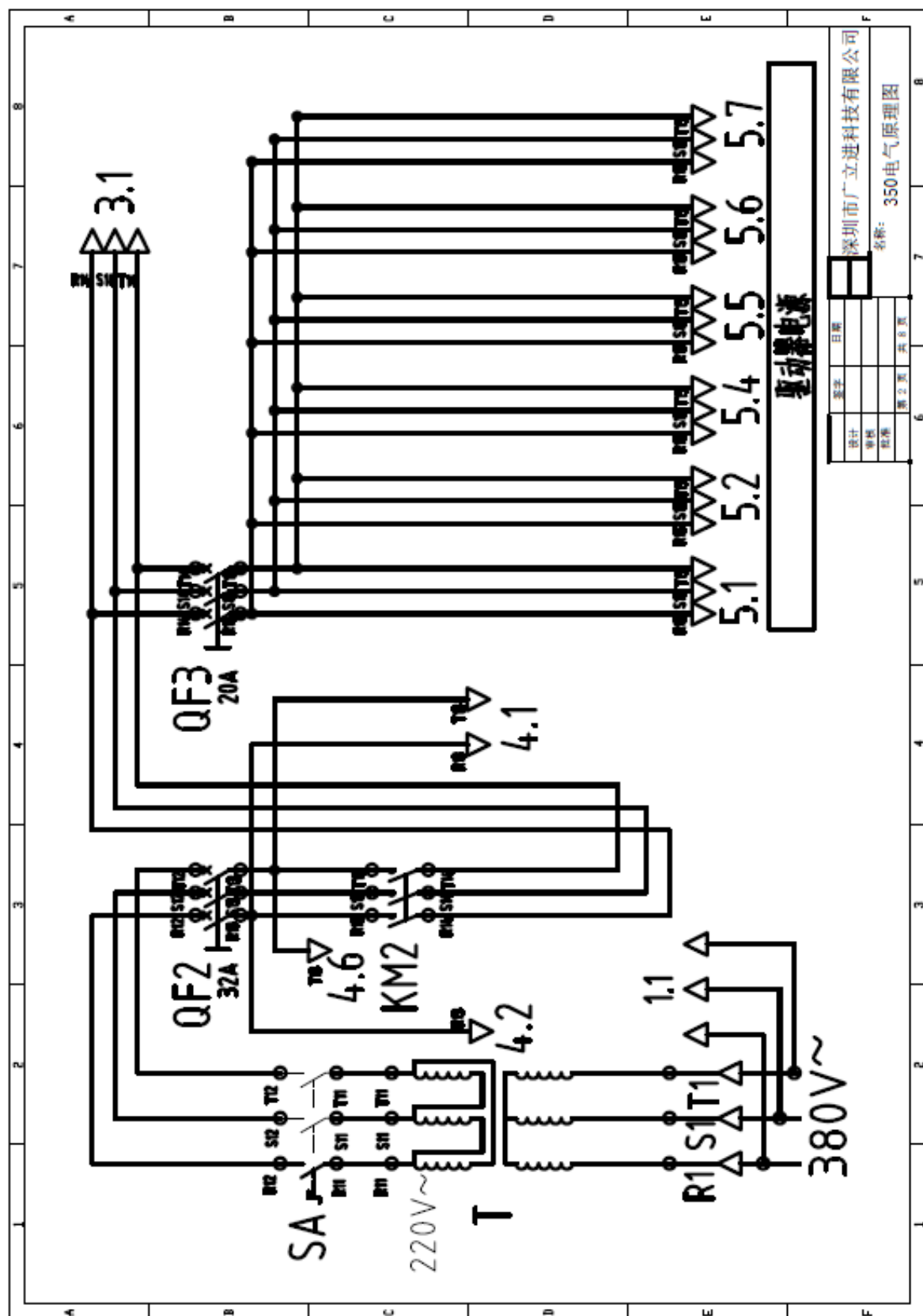
The standard accessories of the machine tool are shown in the table below:

序号	名称及规格	单位	数量	备注
随机附件				
1	水冷机	台	1	
2	刀把 IS020	个	15	
3	扳手 IS020	套	1	
4	六角匙	套	1	
5	喷嘴：ER16 ϕ 3	个	2	
	喷嘴：ER16 ϕ 4	个	2	
	喷嘴：ER16 ϕ 6	个	9	
	喷嘴：ER16 ϕ 8	个	2	
6	油泵	个	1	油箱/油泵
	新油箱	个	1	
	油位尺	个	1	
	单向阀及弯头	个	1	
	铝接头	个	1	
	喉箍（10-16）	个	2	
	宝塔	个	2	
7	铝板	个	1	三轴夹具
	铝柱（L=25）	个	4	
	螺丝（M6*40）	个	4	
8	撑嘴（戒指夹具）	个	1	
9	水晶头	个	6	
10	网线	米	20	
11	对刀仪	个	1	
12	变压器	台	1	
13	新代系统功能键及M码	份	1	
14	新代操作及程式手册	套	1	
附加配件				
1	风枪	个	1	赠送
2	弹簧气管	个	1	赠送
3	防冻液	桶	4	赠送
4	切削油 340G	桶	1	赠送
5	底盘	个	1	赠送
6	油雾分离器	套	1	赠送

9. Electrical schematic diagram







设计	日期	签字	日期
审核			
批准			

深圳市广立进科技有限公司

名称: 350电气原理图

第 1 页	共 1 页
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